6. SOME REMARKS

Two referees provided me with useful comments and some questions. The following is a specific answer to their questions: (i) I have not used symmetries, (ii) I have not used the fact that Kings may not be adjacent to each other, and (iii) I have made available the programs through the Editor-in-Chief; these are also available from chinch@me.iitb.ernet.in.

7. EDITOR'S NOTE

Although the author assumed that someone has worked on this topic already, he did not know any reference. In *Schach und Zahl* p. 10, I found some numbers and references, mostly from N. Petrović. The main statement is that positions with 28 pieces have the largest number of possibilities, especially since 4 Pawns having captured 4 other Pawns may allow the remaining 12 Pawns to promote. According to Petrović the number of possibly legitimate positions then amounts 2×10^{43}.

Because the number of positions with 29, 27, 26 and fewer pieces differs some factor of 10^p (with $p \geq 2$) for each number of pieces, it is assumed in *Schach und Zahl* that the upper bound reads $C \times 10^{43}$. Readers are invited to contact Mr. Chinchalkar or to specify their claim in an equally short note.

8. REFERENCE

Bonsdorff, E., Fabel, K., and Riihimaa, O. (1978). *Schach und Zahl*. 3rd Edition. Walter Rau Verlag, Düsseldorf, Germany.

LITERATURE RECEIVED

PERCEPTION AND MEMORY IN CHESS

Studies in the Heuristics of the Professional Eye

A.D. de Groot[1] *and F. Gobet*[2]
with R.W. Jongman

1996 15.7 × 23.9 cm. XX + 316 pp.
ISBN 90 232 2949 5 stitched Dfl. 79.50

Below, we reproduce parts of the authors' preface.
"The history of this book spans several decades, with long hiatuses between periods of activity. The theoretical basis of some of the book, particularly the sections in which problem-solving issues are discussed, is largely due to the first author and goes back to his thesis (1946; translated into English in 1965). In this work, he also introduced the experimental method of asking subjects to recall positions presented briefly. Although a number of interesting findings on masters' perception and memory were documented in that study (see Chapter 1), the results served primarily as evidence of Masters' superior playing strength – attributed to their *knowledge and experience* rather than to their *computational superiority*. It was only after a gap of almost two decades that the badly needed replication – now focusing

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on perception and memory – was taken up in the project *Denken en geheugen I*[1], directed by the senior author and Nico H. Frijda (University of Amsterdam, 1963-1967). The main results are documented in Riekent Jongman's Dutch thesis (1968); however, data from recall experiments with eye-movement registrations, also collected in the context of the project, were not analyzed and discussed in this study. This was because, his grant having run out, Jongman accepted a full-time appointment at the Groningen University, in a very different field indeed – criminology. (...)

"In 1991, the junior author of the book was completing his Ph.D. dissertation at Carnegie-Mellon University, in Pittsburgh. One of the contributions of his thesis was the development of a computer program simulating human behavior in chess perception and memory. In order to test the plausibility of his model, Gobet was looking for chess players' eye-movement protocols, of which only a handful could be found in the literature. Having come across an RITP memorandum written in 1973 by De Groot and Jongman, he was aware that they had collected such data. A visit to Groningen was set up in order to peruse the material, which was too bulky to sent by mail. The eye-movement records and the verbal protocols accompanying them, unearthed from De Groot's archives, were exactly what Gobet was after – except that they were in Dutch and only partly analyzed. The material was not perfect, but no similar eye-movement data were available in the scientific literature. It appeared that this problem could be overcome, thus enabling Gobet to continue his investigations into the nature of chess skill. It was agreed that Gobet would use the eye-movement material, maintain contact with De Groot, and perhaps collaborate with him on a future project. Contact was maintained primarily by correspondence, but also by Gobet's yearly visit to Groningen. (...)

"Joint authorship of the book implies a shared responsibility for the entire text. Both authors contributed equally to this Preface, to Chapters 6, 7 and 9. Chapters 1 through 4 were written primarily by the senior author, Chapters 5 and 8 by the junior author.
The statement "with Riekent W. Jongman", on the title page acknowledges the fact that all data, and many of the ideas, revisited in the book's analytic chapters (2, 3, 4, 6, and 7) stem from his work. It also recognizes the fact that Jongmans, having retired from his full professorship in criminology, again became a team member at the stage of producing the book. With a renewed enthusiasm for the subject matter, he assumed responsibility for the production of the photographs and other figures in the book – in cooperation with the staff of Van Gorcum, the firm that twenty-seven years earlier had published his Dutch dissertation! (...)

"The first chapter sets the stage with an account of the shift in focus from studying thought (De Groot, 1946) to perception (Jongman, 1968). Obviously, both are important subdomains of cognition, but in a substantive area they are not clearly separable. By pointing to ambiguity in the uses of terms such as "perception", and of "seeing" in particular, the boundary with problem solving is shown to be problematic – an issue that has been addressed in later analyses. Next, recapitulation of the main findings and concepts of the original study articulates the need for focusing on perception, and some of the requirements for doing so.

"In Chapter 2, the major question is: How exceptional are the recall achievements of masters in terms of information transfer? The question is converted into the problem of estimating the size of the space of all "master-game-likely" chess positions. It is answered with a step-by-step approximation of this space by means of various statistical and experimental procedures and by techniques originated in information theory. The resulting estimate, an upper limit of 50 bits transmitted within five seconds, is no longer considered "amazing". In a sense, it is "nothing special" – for anyone who has acquired an adequate coding system. However, even if this necessary condition of chess expertise could be met by a not particularly gifted chessplayer, acquiring such a coding system apparently would require years of training and study.

"Chapter 3 begins with a description of the set-up in the main series of the Jongman study. The program consisted of two parts: eighty *recall* experiments – eight Masters and eight Weaker players as subjects in five experiments each – and an equal number of experiments in *position construction*, where the task was to guess the piece location without the benefit of initial information. Each experiment consisted of a series of

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[1] Thought and memory.

trials, that were continued up to the perfect reconstruction of the position. In this chapter, the quantitative results are summarized and analyzed, first by comparing group results, M-subjects (Masters) versus W-subjects (Weaker players), and then by comparing individual results. Measured in various ways – first trial "hit" scores; information transmission per individual chessman; deficit-scores (the latter two based on the innovation of a series of position trials) – the superiority in recall achievements of M over W was large and absolute: there was no overlap, except in one case (out of eighty) where a Weaker player beat a Master. In position construction, there was no significant difference between M and W in the first trial, but in subsequent trials, the Masters were able to build up a considerable advantage. Finally, an exploratory analysis of differences within the Master group leads to a few surprising findings.

"Chapter 4 is devoted to qualitative analyses of the sixteen subjects' retrospective reports. Starting off with the description and conceptualization of a number of remarkable phenomena, the analysis proceeds with an attempt to develop a model of the Master subject's perceptual search around the board, in line with a problem-solving conception of the whole process of perceiving, retaining, and reproducing a position. This analysis results in a list of types of mental activities – "methods", broadly conceived – which are assumed to be used consecutively by Masters during (and after) the five-second exposure of a position.

"Interpretation of retrospective protocol, itself remarkable for its integration of "dynamic analysis", shows how the list can be used – with allowances for skipping of returning to activities. Finally, the findings of the entire chapter are evaluated: retrospectively, as a replication of qualitative findings in the original study, and prospectively, in terms of recommendations for follow-up studies.
Historically, the work done in the project "Denken en geheugen" represents the bulk of the scientific work on chess psychology between the late forties and the early seventies, when Chase and Simon's work in this field attracted the attention of cognitive psychologists, and inspired many experimental studies.

"Chapter 5 discusses these studies as well as theoretical trends that have influenced chess research in psychology since 1973 and argues that many questions that have vexed the field could have been better answered with a more systematic use of verbal protocols and with the analysis of eye movements. Chapters 6 and 7 demonstrate the benefits of such an approach.

"Chapter 6 reports the way in which the eye-movement data of the *Denken en geheugen* project were processed. Having been collected back in 1967-68, with a cunning but comparatively primitive technique, the raw data could not be made interpretable solely by tabulating results and drawing eye-movement patterns. It also required the correction of apparent displacements of whole patterns due to instrumental inaccuracies. In this chapter, a few more general methodological aspects are also discussed, and quantitative results – concerning variables such as the number and duration of fixations or the importance of the squares fixated – are presented. In general, eye-movement variables show inportant differences between Masters and Weaker players.

"In Chapter 7, the eye-fixation records are compared to subjects' retrospective reports on their eye walk. Combining the techniques of eye-movement registration and retrospective protocol analysis raises new problems of interpretation, but also provides us with valuable information on how much, and what, Master subjects remembered about their perceptual processes during the five-second presentation of a position. Moreover, the combined analysis enables us to evaluate: the validity of hypotheses regarding what attracts the subject's attention, how a position is perceived, and how search is organized.

"In Chapter 8, an information-processing model of chessplayers' perception and memory is presented. It is based on the idea that, through their practice and study of the game, chess experts build up a large database of chunks (patterns of pieces). A novelty of this model, in comparison to others, is that it incorporates a simulation of the human learning process in distinguishing frequent configurations (chunks), and in treating these as perceptual units. This model, implemented as a computer program, successfully simulates several important aspects of chessplayers' behavior. The discussion of simulations will be focused on eye-movements data.

"Finally, Chapter 9 presents a discussion between the two authors on a number of possibly controversial issues – minor ones, having to do with data interpretation, as well as fundamental, theoretical, and methodological issues. This chapter will speak for itself."